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An Integrated Supply Chain Model for Predicting Demand and Supply and Optimizing Blood Distribution

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Abstract: *Background*: The blood supply chain (BSC) is crucial for providing safe and sufficient blood, but it faces numerous challenges and needs to be robust and resilient. This study provides a comprehensive model for managing and optimizing the BSC in real-world scenarios, including emergency and routine circumstances and with consideration of health equity concepts. *Method*: Classic time-series models are applied to predict future supply chain circumstances, addressing uncertainty in blood demand and the need for timely supply. A structured framework and medical preferences are prioritized to optimize distribution, minimize blood shortages, minimize wastage due to expiry, and maximize blood freshness. Genetic algorithms (GA) and particle swarm optimization (PSO) are used to solve mathematical models quickly and efficiently, ensuring reliable operation. *Result*: The model's outcomes can effectively meet the daily needs of the BSC and assist decision-makers managing blood inventory and distribution, improving robustness and resilience. *Conclusions*: Utilizing weights allows for the effective management of each objective function to convert the model into a single-objective mixed-integer linear programming (SO-MILP) based on unique conditions, enabling the system to self-adjust for optimal performance, boosting the sustainability of the blood supply chain, and promoting the principle of health equity under diverse real-world settings.

Keywords: blood supply chain; forecasting; resilient; optimization; data driven

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1. Introduction

Blood transfusion is an essential component of the healthcare system, which saves millions of lives each year worldwide. Despite promising results, researchers have not yet been able to produce artificial replacements that can reliably replace the necessity for donated human blood. Hence, the usage of blood and blood components remains necessary for the treatment of individuals diagnosed with cancer, blood disorders, trauma, and critical health issues. While the need for blood is universal, millions of patients requiring transfusion do not have timely access to safe blood, and there is also a major imbalance between the developing and high-income countries in access to safe blood [1].

The blood supply chain (BSC) controls the distribution of blood products from donors to patients at five different tiers: donors, mobile collection sites (CS), blood centers (BC), demand nodes, and patients. Demand nodes may be hospitals, clinics, or other transfusion points. Mobile CSs, BCs, and demand nodes must be coordinated to perform the six main processes associated with blood donation: collection, testing, component processing, storage, distribution, and transfusion [2]. The blood supply chain (BSC) consistently confronts two principal challenges: the unpredictability of blood supply and demand and the limited shelf life and perishable nature of blood and its components. The existence of uncertainty challenges the planning and distribution of essential blood supplies in the first phase, necessitating meticulous planning to prevent shortages within the system. The finite shelf life of blood constrains its storage in the blood banks.

Many studies, including [3–5], have reported the importance of blood freshness for some treatments. For organ transplant patients, platelet demand is classified as an emergency, requiring the acquisition of the youngest available blood platelets. The youngest platelets are preferred for oncology patients, whereas there is no preference regarding the age of the blood platelets for traumatology patients [6]. The transfusion of the oldest red blood cells affects both the short-term and long-term survival rates in cardiac patients, while giving out the youngest available blood platelets is also significant [3]. Conversely, some surgeries, such as cosmetic procedures, are voluntary, adding a new dimension to the classification of patients. To ensure fairness among patients, it is essential to prioritize patients and the requests concerning the blood supply chain issues.

The provision of blood supply mostly relies on voluntary blood donation, making the management of blood inventory and distribution very important. The current continuous blood donation index, reflecting the proportion of persons who have donated blood at least once in the last year, surpasses 54%. Therefore, we can deduce that categorizing and disseminating critical calls based on current circumstances could potentially act as a tactic to tackle urgent crises and disruptions. Reviewing scientific databases reveals that no study has yet addressed this topic.

We conclude that a forecasting and optimization model for the blood supply chain is imperative at all levels, given the significance of blood supply chain management, the influence and efficacy of resilience, and a review of the pertinent literature and research. This model must possess the capability to predict and optimize the supply chain under standard conditions while ensuring resilience in critical scenarios, all within a framework that prioritizes health equity in real-world contexts.

Additionally, the increase in the average age of the population and improvements in healthcare facilities have heightened the need for blood. The provision of adequate healthy blood in compliance with AABB (Association for the Advancement of Blood & Biotherapies) criteria is a fundamental responsibility of the healthcare system and a primary function of Blood Service Centers in both developed and developing countries.

A review of the literature reveals that models are developed for regular or emergency conditions, with no model relevant to all possible circumstances. The integration of forecasting tools to enhance the preparation of the BSCs for many potential situations has not yet been accomplished. Because of this, we have decided to make a complete model that can handle all possible outcomes within the given framework, along with a forecasting model to predict what will happen and maintain its resilience. As the concept of equity gains importance, it is essential to address this critical issue within the healthcare system, particularly in the blood supply chain, which is a vital component of healthcare with no artificial alternatives.

This study aims to develop an integrated, resilient, and data-driven supply chain model for the blood distribution network. The proposed model will predict supply and demand levels for various blood groups during designated planning intervals. Simultaneously, it will optimize blood distribution and supply chain goals to minimize blood shortage levels, minimize wastage due to expiration, and maximize freshness, thereby ensuring resilient performance under a variety of real-world scenarios. Unlike the existing studies, this work leverages innovative data-driven methodologies to create a practical optimization model tailored for real-time implementation. This approach contributes to improving the overall effectiveness and efficiency of BSC management, aligning with the goals of ensuring healthcare resilience and sustainability. This technique builds upon previous studies by employing a scenario selection algorithm, which prioritizes patients according to their urgency level, thereby promoting health equity in atypical situations. Therefore, we assign weights to each scenario to prioritize its impact on the model.

This paper aims to address the following questions: What is the impact of incorporating forecasting into the optimization processes' overall performance? How can we employ strategies to create an optimal blood distribution plan that takes into account the urgency of demand and the current inventory levels? What is the optimal approach for blood allocation considering the demand urgency and current inventory levels? What strategies can we employ to optimize blood distribution in various scenarios? How can we design and optimize the blood supply chain to ensure resilience against critical situations, shortages, and fluctuations in supply and demand? Which strategies does the system implement to address the challenges posed by real-world environments and emergencies, with the aim of reducing risks to patient safety? How do supporting facilities affect the blood supply chain's performance, and how can they enhance model optimization?

We structured this study based on prior research in the literature about the forecasting and optimization of BSC, which we thoroughly discuss in the Section 2, along with observations and interviews with medical and transfusion specialists at IBTO, Tehran Branch. We collected the data necessary to validate the proposed model from the IBTO database, which includes data on blood supply, demand from medical facilities, blood allocation compared with demand, and the wastage ratio due to expiration from March 2017 to January 2022.

We have developed a systematic procedure to achieve the structure and objectives: In the initial step, we forecast the future status of blood demand and supply using the time-series models. We then elucidate the status of the system by comparing the forecasted output from the initial step with the actual input values of the system and using a scenario selection flowchart (detailed in Section 3). We develop the multi-objective optimization model in the final step to minimize blood shortage, reduce blood wastage due to expiration, and maximize blood freshness. We transform the model into a single objective optimization problem by applying weights to each objective function, as defined in the second step of the procedure. The weights of the objective functions vary in each scenario due to the differing conditions of the model. This method and procedure can effectively address various situations. This method utilizes an optimization model grounded in compiled allocation policies to uphold health equity among diverse patients, thereby establishing an optimal front. This procedure enables decision-makers within the system to prioritize goals in order to attain the desired optimality.

2. Literature Review

The study of the regional and local management features of blood supply began mostly in the 1960s, peaked in the late 1970s and early 1980s, and then dropped off significantly to the present time [7]. Since 2017, the blood supply chain has again been the focus of researchers, and there has been a trend to increase the number of studies in the field. During the last four years, prestigious journals have published over 120 articles in this field [8]. We examined the literature in two independent sections to encompass the forecasting and optimization of the blood supply chain.

2.1. Forecasting Methods Used in Blood Supply Chain

The increasing popularity of BSC has captured the interest of researchers, leading to a focus on investigating the forecast methods and algorithms and their implementation in BSC. The authors of [9] created time-series and machine learning algorithms to analyze five years of historical data on the supply of blood from the Taiwan Blood Services Foundation; the most efficient approach was determined. The autoregressive (AUTOREG), autoregressive moving average (ARMA), autoregressive integrated moving average (ARIMA), seasonal ARIMA, seasonal exponential smoothing method (ESM), and Holt–Winters models were used to look at the time series. Machine learning algorithms classify both artificial neural networks (ANN) and multiple regression. The seasonal ESM and ARIMA models gave the fewest error measures when compared to other approaches, demonstrating the superior performance of time-series approaches over machine learning algorithms. The authors of [10] developed an adaptive evolutionary support vector regression model for forecasting the emergency blood demand for the day after an earthquake. They based their research on data from the 2008 Wenchuan earthquake in China. The performance of the created model was compared to forecast the demand for red blood cells, plasma, and whole blood. In addition, other models such as ARIMA, SES, DES, TES, SMA, EMA, and MLR have been evaluated by calculating the forecasting error of each model. The developed AEVSR model outperformed other models, including the ARIMA and regression approaches, according to the results.

A research effort conducted by [11] reviewed the current literature on blood demand forecasting. The study sought to identify the criteria for assessing operational and tactical level forecasts, as well as to review the forecasting methodologies currently employed by the Finnish Red Cross Blood Service. The ARIMAX and combination forecast (CF) models provided superior performance compared to all other models when predicting the RBC series, with a significant margin. Other superior candidates include a TBATS model, an ETS model, and perhaps surprisingly, the seasonal naive model. Simple moving-average models typically drive more complex models when forecasting by blood type in series with smaller volumes. The ARIMAX and seasonal naive models demonstrated superior accuracy compared to other models.

In ref. [12], a mathematical model was proposed to address the disparity between excess and inadequate blood units in blood banks across three primary stages. The model aims to forecast the demand for blood units at the blood bank, optimize the distribution of units from surplus blood banks to those with shortages, and identify the most efficient route for delivering the allocations. The forecasting procedure utilizes the ARIMA, seasonal ARIMA, and exponential methods for forecasting demand. A monthly sample dataset spanning 25 years was utilized to establish the model. Subsequently, forecasts were generated for the following 24 months. The findings revealed that SARIMA outperformed the exponential techniques for the time periods of approximately 1 year. However, for time periods beyond two years, exponential approaches appear to yield similar or even superior outcomes in certain instances. The study conducted by [13] utilized data from March 2018 to April 2019 to forecast the blood collection in a major hospital in China. The researchers employed ARIMA, seasonal ARIMA, linear regression methods, and a combination model to analyze the data. They also implemented a weighted average method to adjust the residuals of two sets of individual models. The results demonstrated that the seasonal ARIMA model is proficient at accurately forecasting with minimal error. Conversely, the combined model produces precise forecasts and offers a novel approach to forecasting the demand for the blood collection service in the outpatient department.

The prediction of blood demand for all blood groups in the city of Shiraz from 2012 to 2019 was conducted using three models: ARIMA, ANN, and a merged model of both ARIMA models. The results demonstrate that the ARIMA model performs well in forecasting applications for a 12-month period. Additionally, they utilized the research findings to demonstrate a declining pattern in the prevalence of positive blood groups and an increasing pattern in the prevalence of negative blood groups [14].

Authors of [15] utilized statistical modeling, machine learning, and optimization techniques to create a data-driven approach for forecasting current demand and managing it effectively. They utilized their newly integrated model to forecast the demand for Canadian hospitals and optimize the blood supply management for a duration of two weeks, with the aim of minimizing blood circulation. They developed a demand forecast model that reliably correlates with the current weekly optimization model by employing machine-learning techniques like LSTM and XGBoost. The outcomes of their efforts highlight the proficient execution of the constructed model and its capability to make predictions with tolerable margins of error.

2.2. Optimization Methods Used in Blood Supply Chain

The majority of literature on the blood supply chain concentrates on optimizing its operations, whether in normal or disastrous conditions. Ref. [16] Created a comprehensive model that combines simulation and optimization techniques to assist in making strategic and operational decisions on blood production planning. The appropriate linear number optimization model is applied to the planning horizons to facilitate daily scheduling, which includes determining the necessary donor count, bleeding procedure, and production planning. Blood center managers can utilize their model in two distinct modes. The first mode enables the blood center managers to assess various resource allocation plans at a strategic level. The second mode, on the other hand, establishes daily collection and production objectives at an operational level.

A mathematical model was presented by [17] to regulate the existing blood in a hospital, considering unknown variables and utilizing chance constraints in programming. The model considered the uncertain demand for blood from the central blood bank. It divided the requested blood into fresh blood and normal blood based on the patients' conditions. The model aimed to minimize the rate of blood deficiency and degradation by employing an appropriate blood group replacement method. They proposed that other medical facilities could supply contaminated blood at a lower cost than fresh blood. The findings indicate that decreasing the transfer rate resulted in a decrease in blood clotting within the hospital. Additionally, increasing the blood replacement rate led to a further decrease in shortages and wastage of blood during the planned time.

A stochastic bi-objective model of the blood supply chain was implemented by Fahimnia et al. [18] to optimize efficiency and effectiveness, with the objective of responding promptly during times of epidemics and emergencies. Their strategy is to minimize the expenses associated with the whole supply chain and decrease the delivery time to hospitals. In order to solve the problem, they employed a technique that involved integrating the $\in$-constraint with the Lagrangian relaxation. Ensafian et al. [19] examined four models of mixed integer programming, specifically focusing on the FIFO and LIFO approaches to random number planning. The objective of these models was to maximize blood freshness and reduce system expenses. They utilized stochastic planning to account for uncertainty, and they employed goal programming, robust optimization, and multi-objective goal programming to solve the four suggested models. The performance of each of the proposed models was assessed by analyzing the data from the Tehran Province blood transfusion network. In ref. [20], a bi-objective integer programming model integrates the collection, processing, production, distribution, and vehicle routing of blood supplies. The model aims to optimize both the costs and freshness of the blood transferred to hospitals. The researchers employed a multi-stage random planning approach and integrated it with a scenario tree to compensate for uncertain conditions. To address the complexity of the matter, the researchers developed a self-adapted multi-objective-combined differential evolutionary algorithm. This method utilizes fuzzy-dominant ordering to search for variable neighbors. The method's performance is evaluated by comparing it with two widely used multi-objective evolutionary algorithms, MOICA and NSGA-II.

Ref. [21] presented a single-objective nonlinear mixed-integer programming model to minimize unfulfilled requests in critical scenarios. They utilized cross-match methods and blood group matching to allocate blood to the requests. To address the blood allocation system in critical circumstances and provide relief, they suggested using a greedy heuristic approach to reach a near-optimal solution. Diabat et al. [22] proposed a framework for enhancing the efficiency of the blood supply chain, considering the four tiers that generally exist during emergency scenarios. They specifically developed their bi-objective model, a resilient stochastic planning model, to reduce delivery time and overall cost in real-world situations characterized by uncertainty, particularly during crises. They solved the model using Mashhad city data and the technique of e-constrained and Lagrangian relaxation, taking into account diverse variables like disruptions in the blood supply system.

A paper by [23] proposed a novel methodology for addressing the stochastic optimization models in extensive blood supply systems, incorporating machine learning methods. The researchers employed various machine learning algorithms, such as artificial neural networks, classification and regression trees, K-nearest neighbors, and random forests, to create a model that could make efficient operational decisions. The researchers observed that while the machine learning algorithms perform well and achieve near-optimal results, they cannot outperform precise algorithms and cannot replace them.

Alizadeh et al. [24] developed a multi-level bi-objective model to manage the cost and duration of blood transfer in the four time periods following an earthquake: the first hour, the second hour to twenty-four hour, the third hour, and the third hour till the end of the first week. In their model, they assume a Babol city earthquake and blood demand within certain timeframes. They propose establishing mobile bases to meet the blood demands while minimizing costs and delivery time. They employ an approach that involves transforming a multi-objective model into several single-objective models, then establishing a trade-off point to determine the optimal solution. The model-solving approach requires establishing an order of objectives for the problem before initiating the solution. The struggle between objectives results in an uncertain level of optimism, and as priorities change, the level of optimization likewise varies, leading to the building of the Pareto optimal front.

The authors of [25] presented an approach to modeling the UK's blood platelet demand. They employed a novel simulation technique that serves as a decision-support system to ascertain the optimal amount of blood platelets needed for critical circumstances, with the goal of ensuring availability and minimizing wastage. The primary goal was to develop a model that can determine the ideal quantity of blood platelets needed to meet requests and minimize shortages. The model also considers the impact of hospital demands and supply dependency in order to determine the appropriate stock level and provide a trustworthy system for crisis circumstances.

The study conducted by [26] in the field of blood supply chain research focused on prioritizing patients in order to optimize the allocation of blood banks. The treatment centers' requests were classified into two overarching categories: "elective" and "non-elective". They further prioritized them into three levels: "high", "medium", and "low". In addition, they accounted for two variables: the freshness of the blood and the possibility of using compatible blood types to distribute blood upon request. The given model does not address the subsequent priority until it fulfills the requirements of the preceding priority. The system prioritizes an entity by using new blood of the same blood type upon demand. The second priority involves a mixture of both fresh and ordinary blood, as well as blood groups that are compatible with one another. Assignments to third-priority applications do not consider fresh blood status. However, they encourage the use of fresh blood and allow compatible blood groups. They design their priority and blood allocation policies to prevent shortages, particularly for the first group, which prioritizes emergency patients and those with serious illnesses. This strategy is the sole instance of applying the concept of a health equity system in the management of blood inventory. In order to optimize the efficiency of blood transfer to treatment centers and minimize blood shortages, optimization techniques were employed in a model presented by [27]. This involved using modeling to simulate uncertain demand and supply scenarios and then optimizing target functions based on the results of these simulations. Requests for the allocation of three priority groups were assessed based on the patients' conditions. Requests were prioritized and addressed accordingly. In addition, for requests with the lowest priority, the option of employing replacement blood groups was taken into account. A sensitivity analysis was performed on the affirmations by evaluating several adjustment rules, such as FIFO and LIFO.

In ref. [28], the authors proposed a probability model for two stages of a focused stable scenario, taking into account the occurrence of two earthquakes and pandemics in the multi-echelon blood supply chain of Tehran Province and their consequences. They created six scenarios to replicate possible disruptions, reduce the expenses of establishing fixed and mobile facilities, transportation, and procurement of essential equipment and transportation, minimize the time delay between consumption and production, and minimize shortages by employing a resilient strategy. They demonstrated that they can optimize the blood supply system and minimize damage in the presence of diseases by utilizing surplus capacity, enhancing cognitive flexibility, and promoting social responsibility.

The authors of [29] considered an earthquake in Istanbul; a precise numerical model was created that combined multiple time frames and bi-objectives inside a multi-scenario framework. The goal of this model was to minimize both the unfulfilled need for blood and the overall cost of the network. In their model, they also explored the potential of utilizing compatible blood groups to address requests from various blood groups, as well as the categorization of patients in hospitals according to their vulnerability to disasters (high priority and low priority). Furthermore, it is important to analyze the current conditions following the earthquake, including the extent of damage to the centers and how it affects the performance of each. The recommended solution, presented in a timely manner, employed the techniques of the ε-constraint method and Lagrangian relaxation.

2.3. Research Gap

We have separately reviewed the existing literature regarding forecasting methodologies and optimization models employed in BSC. Table 1 summarizes the literature review, presenting an overview of the models and methodologies and comparing them.

This summary delineates the forecasting methodologies employed in publications relevant to the forecasting literature section. Furthermore, we have classified the mathematical models and objective functions of each study to delineate their specific objectives. The literature review and summary table illuminate the modeling and decision-making contexts at different levels of the BSC, offering a more precise comprehension of the modeling terminology utilized in each work. Furthermore, studies have examined the implementation of resiliency and health equity principles, highlighting the models' dependability and effectiveness in practical situations. We examine the cross-matching of blood groups to comprehend how models integrate medical principles.

The literature examines the lifespan of each blood component, which affects the model, and evaluates the models using a case study. This assessment is essential for confirming that the models not only embody theoretical potential but also exhibit practical efficacy across many contexts. Researchers seek to refine models by analyzing these case studies and integrating discoveries that improve their relevance in practical medical contexts.

The literature review reveals that not all publications are applicable in all situations. Instead, the focus is on optimization models specifically designed for either normal or catastrophic situations. Additionally, a manuscript outlining the amalgamation of a forecasting model and an optimization model to improve the efficacy of the blood supply chain under diverse system circumstances has not undergone peer review.

We conducted this study to fill the void in the literature and set ourselves apart from earlier research by presenting a comprehensive forecasting and optimization model that can adapt to real-life situations. We achieve this by employing a scenario selection model and implementing predetermined actions based on the model's diagnosis. The proposed model's case study provides the basis for these actions, which aim to ensure safe and sufficient blood assignment.

In the assignment policy, health equity is taken into account to prioritize demands, along with the substitution of blood groups through cross-matching. The proposed mode utilizes an order to assign fresh blood based on urgency levels, ensuring proper management of the blood inventory and blood assignment. Altogether, the proposed integrated model will contribute to being a practical, resilient optimization model that will manage the blood distribution network and also manage demands from the backup/supporting facilities that are planned, allowing them to communicate with each other. This approach differentiates this study from prior work, addressing critical gaps in the literature.

Table 1. Summary of literature review on forecasting and optimization methods in the blood supply chain.

Reference No.	Mathematical Model	Objective Function				State environment	Supply Chain Echelons	Decision-Making Environment	Use of Resiliency Concept	Forecasting	Prioritizing Demands	Machine Learning	Cross-Matching	Blood Component	Case Study
		Cost	Shortage	Wastage	Blood Freshness										
1	N/A	-	-	-	-	-	Multi	Deterministic	-	-	-	-	-	Whole Blood	Iran
2	N/A	-	-	-	-	-	-	-	-	-	-	-	-	-	Systematic Review
3	Multi-objective		✔	✔	-	Normal	Single	Deterministic	-	-	-	-	-	Whole Blood	Chicago, USA
4	N/A	-	-	-	-	-	-	-	-	✔	-	✔	-	-	-
5	-	-	-	-	-	-	Single	-	-	Supply	-	✔	-	RBC	Taiwan
6	-	-	-	-	-	-	-	-	-	Demand	-	-	-	-	Earthquake, China

Table 1. *Cont.*

Reference No.	Mathematical Model	Objective Function				State environment	Supply Chain Echelons	Decision-Making Environment	Use of Resiliency Concept	Forecasting	Prioritizing Demands	Machine Learning	Cross-Matching	Blood Component	Case Study
		Cost	Shortage	Wastage	Blood Freshness										
7	-	-	-	-	-	-	-	-	-	Demand	-	-	-	RBC	-
8	MIP	-	✔	-	-	Normal	Single	Deterministic	-	-	-	-	-	-	India
9	-	-	-	-	-	-	-	-	-	Supply	-	-	-	-	China
10	-	-	-	-	-	-	-	-	-	Demand	-	-	-	-	Shiraz, Iran
11	-	-	-	-	-	-	-	-	-	Demand	-	✔	-	-	-

Table 1. *Cont.*

Reference No.	Mathematical Model	Objective Function				State environment	Supply Chain Echelons	Decision-Making Environment	Use of Resiliency Concept	Forecasting	Prioritizing Demands	Machine Learning	Cross-Matching	Blood Component	Case Study
		Cost	Shortage	Wastage	Blood Freshness										
12	-	✔	-	-	-	Normal	Single	-	-	-	-	-	-	RBC/PIT/Plasma/Cryo	Columbia
13	Multi-objective IP	-	✔	✔	-	Normal	Single	Probabilistic	-	-	-	-	-	-	-
14	Two-Stage SP	✔	-	-	-	Critical	Multi	Stochastic	-	-	-	-	-	-	-
15	Robust Stochastic MIP	✔	-	-	✔	Normal		Stochastic	-	-	-	-	-	PLT	Tehran, Iran
16	Multi-Stage SP	✔	-	-	✔	Normal	Multi	Stochastic	-	-	-	-	-	PLT	Sari, Iran

Table 1. *Cont.*

Reference No.	Mathematical Model	Objective Function				State environment	Supply Chain Echelons	Decision-Making Environment	Use of Resiliency Concept	Forecasting	Prioritizing Demands	Machine Learning	Cross-Matching	Blood Component	Case Study
		Cost	Shortage	Wastage	Blood Freshness										
17	MIP	-	✔	-	-	Critical	Single	Stochastic	-	-	-	-	✔	RBC	-
18	Bi-Objective RO	✔	-	-	✔	Critical	Multi	Stochastic	✔	-	-	-	-	-	-
19	Two-stage SO	✔	-	-	-	-	Multi	-	-	-	-	✔	-	-	-
20	Bi-Objective IP	✔	-	-	✔	Critical	Multi	-	-	-	-	-	✔	RBC	Earthquake, Babol, Iran
21	Multi-objective	-	-	-	-	Critical	Multi	Stochastic	✔	-	-	-	✔	PLT	UK

Table 1. *Cont.*

Reference No.	Mathematical Model	Objective Function				State environment	Supply Chain Echelons	Decision-Making Environment	Use of Resiliency Concept	Forecasting	Prioritizing Demands	Machine Learning	Cross-Matching	Blood Component	Case Study
		Cost	Shortage	Wastage	Blood Freshness										
22	MIP	✔	✔	✔	✔	Normal	Single	Stochastic	-	-	✔	-	✔	RBC	Tehran, Iran
23	MIP	-	✔	-	✔	Critical	Multi	-	-	-	✔	-	✔	-	China Blood Bank
24	Two-Stage Robust Stochastic	✔	✔	-	✔	Critical	Multi	Stochastic	✔	-	-	-	-	RBC/PLT/Plasma	Earthquake, Tehran, Iran
25	Bi-Objective Multi-Period Multi-Scenario	✔	✔	-	-	Critical	Multi	-	-	-	✔	-	✔	-	Earthquake, Istanbul, Turkey

Table 1. *Cont.*

Reference No.	Mathematical Model	Objective Function				State environment	Supply Chain Echelons	Decision-Making Environment	Use of Resiliency Concept	Forecasting	Prioritizing Demands	Machine Learning	Cross-Matching	Blood Component	Case Study
		Cost	Shortage	Wastage	Blood Freshness										
26	Multi-objective	✔	✔	✔	-	Normal	Single	-	-	-	-	-	-	PLT	-
27	Bi-Objective Two-Stage SP	✔	-	-	-	Critical	Multi	Stochastic	-	-	-	-	✔	RBC	Mashahd, Iran
28	Stochastic integer programming	-	✔	✔	-	Normal	Single	Stochastic	-	-	-	-	-	PLT	-
29	Multi-objective	-	✔	✔	-	Normal	Multi	Stochastic	-	-	-	-	-	PLT	Stanford University Medical Center
This Study; Multi-Objective MIP		-	✔	✔	✔	✔	✔	-	✔	-	✔	-	✔	**RBC**	**Tehran, Iran**

3. Materials and Methods

The goal of this research is to optimize the blood supply chain from donors to patients using a general model suitable for all scenarios. In real-life scenarios, fluctuations in blood supply and demand necessitate numerous decisions to optimize the supply chain and meet the objective functions.

To perform the optimization, first, a prediction model is used to predict the required products and prepare the supply chain to satisfy the demands from hospitals and other medical centers. Second, after a comparison between the predicted values and actual values, which will be entered to update the model, and from possible scenarios that were defined in the prior step, an appropriate decision will be made. This procedure will transform the model into a data-driven model suitable for all scenarios. Many methods can address the inherent uncertainty in blood supply and demand data, but live data streaming flows enable data-driven predictions.

Hospitals and blood collection centers employ a time-series analysis and prediction system as a sub-system to forecast their blood requirements. Based on medical experts, blood required varies from time to time and season to season, so time-series analysis is a good and proper tool to make predictions. We will account for supply and demand uncertainty by implementing the time-series-based sub-system.

The first step involves using forecasting models, such as seasonal auto regressive integrated moving average (SARIMA) and auto regressive integrated moving average (ARIMA), to forecast the demand for blood from hospitals and other facilities that treat patients who request blood, as well as the future status of blood supply into the supply chain. To conduct accurate forecasting, historical data for both blood supply and demand are gathered. We then undertake a thorough process of data cleansing to ensure the reliability and integrity of the dataset. We carry out an in-depth analysis following data preparation to determine the optimal parameters for each forecasting model, specifically the parameters p, d, and q for the ARIMA model. These parameters play a crucial role in defining the model's structure and capturing the underlying patterns within the time-series data. The values of p, d, and q show the autoregressive, differencing, and moving average parts of the data, respectively. They show how the data change over time and show trends. This meticulous analysis serves as a vital step in enhancing the forecasting accuracy and reliability of the ARIMA model, ensuring it is well suited to capture the nuances of the blood supply and demand dynamics.

In a similar vein, the forecasting process extends to the utilization of the Seasonal Autoregressive Integrated Moving Average (SARIMA) model. Initial steps involve the compilation of historical data pertaining to both blood supply and demand. We then implement rigorous data cleansing procedures to rectify any anomalies and ensure the integrity of the data. Following the data preparation phase, we conduct an extensive analysis to determine the optimal parameters for the SARIMA model, identified as p, d, q, P, D, Q, and s.

These parameters are essential for capturing the autoregressive, differencing, moving average, and seasonal dynamics within the blood supply and demand time series. By employing an iterative and data-driven approach to parameter selection, we ensure the model adapts effectively to the intricate seasonal patterns and temporal variations characteristic of blood-related variables. This optimization significantly enhances the SARIMA model's predictive accuracy, making it highly suitable for addressing the complexities of real-world blood supply chain scenarios. The comprehensive nature of this parameter tuning process underscores the model's capability to respond to fluctuations in demand and supply, thereby improving its robustness and reliability in diverse contexts.

The selection of the most suitable forecasting model for blood supply and demand is contingent upon a thorough evaluation of forecasting errors. This assessment involves comparing the performance of various models using actual data obtained from the supply chain alongside the predictions generated by each forecast model. The forecasting errors, which may include metrics such as mean squared error (MSE) or root mean squared error (RMSE), serve as quantitative indicators of how well each model aligns with the observed outcomes.

In this study, we selected root mean squared error (RMSE) and mean absolute Error (MAE) as evaluation metrics due to their direct interpretability in the context of blood demand. These metrics are expressed in the same units as the forecasted variable (blood units), making them tangible and actionable for healthcare stakeholders. RMSE was chosen for its sensitivity to larger errors, which are critical to avoid in blood supply chain scenarios, while MAE provides a balanced average error measure. Together, these metrics emphasize minimizing absolute deviations in demand forecasts, aligning with the practical and operational priorities of this study.

We identify the optimal choice as the model that exhibits the least forecasting error, thereby exhibiting greater accuracy in predicting blood supply and demand dynamics. This careful review makes sure that the chosen forecasting model fits well with the complex factors related to blood, which makes it more reliable to guess what will happen with supply and demand in the BSC in the future.

In the subsequent step, the model plays a pivotal role in discerning the prevailing situation—whether it is within normal parameters, transitioning into a semi-crisis, or already in a state of crisis. This determination hinges on a meticulous comparison between the actual data and the forecasted values derived from the earlier step. Leveraging the outputs generated in the initial phase as variables, the model systematically evaluates the concordance between the anticipated and actual figures within a specified confidence interval.

This comparative analysis unfolds in two stages, initially focusing on the demand for blood and subsequently extending to the supply. We use the results of these comparative assessments to build a predefined scenario tree, which we present below. Each potential scenario within this tree is associated with a distinct classification, guiding the system's response to the observed conditions. Once the model identifies a specific scenario, it seamlessly transitions into recommending the necessary actions to maintain the sustainability of the BSC. The model strategically tailors these recommended actions to meet the unique demands of each scenario, ensuring a proactive and adaptive approach to maintain the equilibrium and resilience of the overall system.

Through this comprehensive framework, the model not only facilitates real-time situational awareness but also empowers decision-makers with actionable insights for effective and timely intervention in support of BSC sustainability. Through the implementation of these measures, BSC can strengthen its resilience in addressing the challenges that the model may encounter. Moreover, it establishes the optimization model as the final phase of the integrated model, configuring objective functions to optimize it according to various potential scenarios and the corresponding required actions. We craft the scenario tree by comparing actual and forecasted values, using the latter as the input from the initial step. Moreover, it establishes the optimization model as the final phase of the integrated model, configuring objective functions to optimize it according to various potential scenarios and the corresponding required actions.

In the third step, we develop a multi-objective optimization model to optimize the BSC, taking into account the needs of each selected scenario from the prior step. Medical experts classify demands into three main urgency classifications to establish an equity-oriented BSC. Each period's blood demands are independent and must be satisfied within that same period. The blood in the first half-life is considered fresh, and the second half-life is considered ordinary.

To clarify the concept of the proposed model for this research, a flow chart (Figure 1) consisting of all echelons is prepared as follows. The proposed flowchart illustrates the physical flow of blood from donors to patients, also known as the cold chain, and the data flow that supplies the necessary information for the decision support system.

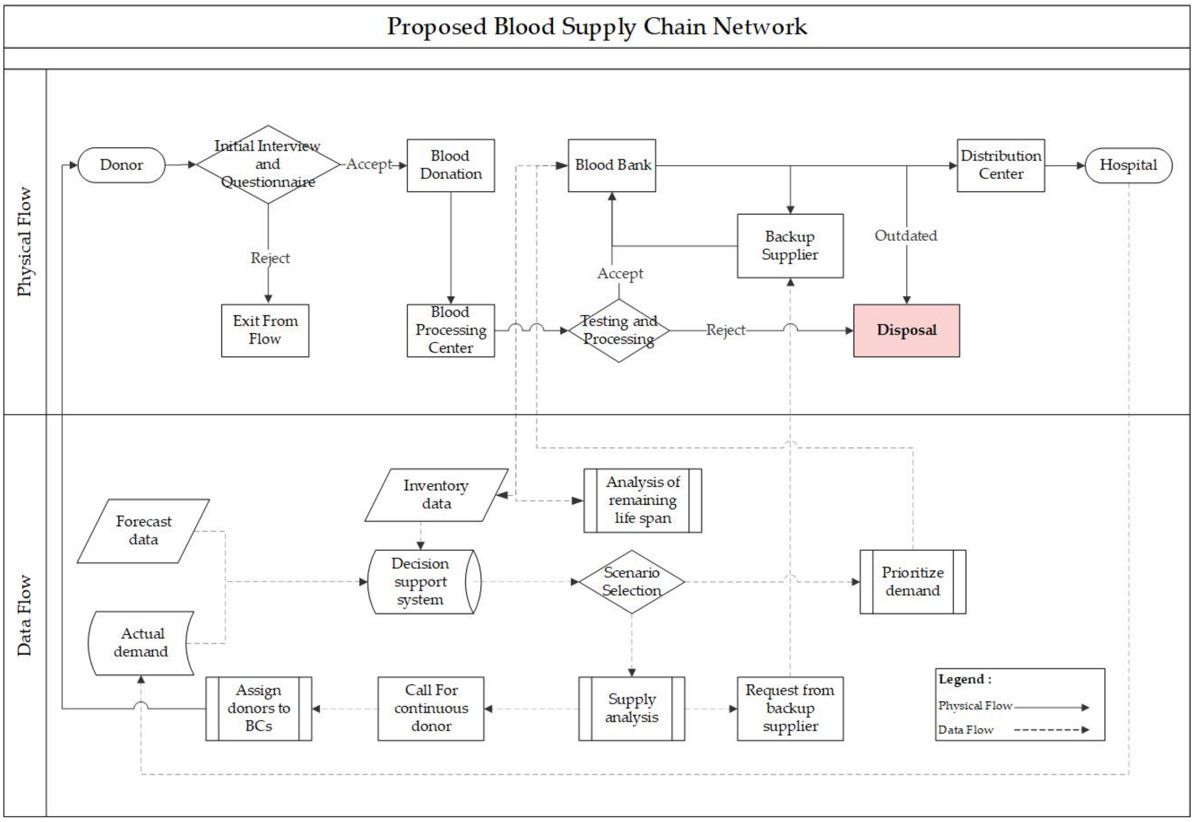

Figure 1. Proposed blood supply chain network.

To take into account health equity for all patients and fulfill the demand based on the urgency classification, a replenishment and issuing policy is considered below:

Class 1, High-Priority: These are patients with life-threatening conditions, such as severe bleeding or organ failure, that may be caused by disease or due to disasters happening. We will give them the highest priority in receiving blood transfusions. Due to the LEFO (Last Expired First Out) issuing policy, the same blood group type in the freshest available form is assigned to satisfy this class of demands. This class does not allow blood shortages.

Class 2, Moderate-Priority: These patients with less serious conditions may require blood transfusions shortly. We will give them priority once we have served the high-priority group. A mix of LEFO and FEFO (first expired, first out) is used to fulfill this type of demand. Firstly, issuing the same blood group in fresh form is considered; if the required inventory is not available, then substitution with the appropriate blood group following the table and priority weights is used to fulfill the demands.

Class 3, Low-Priority: These are patients with less serious conditions who may not require blood transfusions immediately. They will receive blood transfusions last, as more blood becomes available. The FEFO policy applies to this class of demands, and it permits substitution with a blood group that is compatible.

Based on the established prioritization scheme, the processing sequence commences with addressing first-priority requests. If we allocate all the prerequisites associated with first-priority requests, the processing stream smoothly shifts to second-priority requests. Once we have thoroughly processed the second-priority requests, we turn our attention to the third-priority requests.

In the blood substitution and compatibility t between donor and patient table (Table 2), the rows represent the blood type of the patient, and the columns show the blood types that the patient can receive. The table displays the same blood group with a weight of one, compatible blood groups with a weight of two, and incompatible blood groups with an infinity symbol. For example, a patient with type O− blood can only receive blood from a donor with type O−, since type O− blood is considered the "universal donor" and does not contain any antigens that could cause a reaction in the recipient. On the other hand, a patient with type AB+ blood can receive blood from any blood type, since type AB+ blood is considered the "universal recipient" and contains all antigens.

Table 2. Blood compatibility table between donor and patient.

Donor	Patient							
	AB+	**AB−**	**B+**	**B−**	**A+**	**A−**	**O+**	**O−**
AB+	1	∞	∞	∞	∞	∞	∞	∞
AB−	2	1	∞	∞	∞	∞	∞	∞
B+	3	∞	1	∞	∞	∞	∞	∞
B−	3	2	2	1	∞	∞	∞	∞
A+	3	∞	∞	∞	1	∞	∞	∞
A−	3	2	∞	∞	2	1	∞	∞
O+	4	∞	3	∞	3	∞	1	∞
O−	4	3	3	2	3	2	2	1

For administering blood transfusions to patients in need, the possibility of utilizing blood from either a compatible blood group or an identical blood group exists. This is determined through reference to a table compiled during various experiments by researchers and physicians. The table (Table 2) indicates the feasibility of using specific blood groups for transfusions to patients with differing blood groups without encountering adverse reactions. While addressing this matter, priority is assigned based on compatibility between blood groups. When the need arises for the use of compatible blood groups, the appropriate blood group is selected using this table.

Due to an invaluable product that has no alternative to be replaced, having a sustainable and resilient supply chain adds to its importance. This research uses two methods considered to add to the resiliency of systems: using multiple sources and backup suppliers. Continuous and with-history donors, as per the definition presented by blood transfusion experts, are those who gave blood once or more in the last 12 months. More than 60 percent of the required blood is supplied from this group of donors. The significant volume of blood supplied by them adds to the importance of planning the system based on them and having a schedule to normalize the flow of blood in BSC. Another source for supplying blood is backup regions, which are considered backup suppliers of blood whenever an enormous shortage occurs due to tragic situations or a significant rise in donation due to social aspects has occurred.

Prior to delving into the mathematical section of the article, it is essential to provide a conceptual framework of the proposed blood supply chain (Figure 1). This clarifies the data flow and movement within the network, as well as the functionality of the mathematical concept. We define two concurrent processes that are designed to operate sequentially. The data flow begins with demand and supply forecasts, which are derived from a time-series model and are supplemented by actual demand submitted by medical institutions. In parallel with the physical process, blood collection centers (BC) inquire about the lifestyle habits and medical history of donors during an initial interview. Upon completion of this phase, people become qualified to give blood. The blood processing center (BPC) receives donated blood and conducts tests to confirm the absence of diseases. Blood banks segregate all transported blood from blood collections (BCs) to blood processing centers (BPCs) until they disclose test results, allowing for their classification as inventory units. Blood banks dispose of the discarded blood units during testing.

The decision support system receives the inventory data for analysis and selects a suitable scenario. This study aims to minimize the primary issue of expired blood units by recalibrating the remaining lifespan of inventory units. After selecting a scenario, we schedule specific activities for implementation. Prioritizing requests according to patient urgency, requesting blood from backup sources, or appealing for donations from continuous donors are strategies intended for sustaining system performance and resilience during critical circumstances. Transferring excess inventory units to backup facilities is likely to avoid expiration.

Subsequently, the decision support system discloses its result, executes requisite actions in accordance with system rules to fulfill its objectives, and allocates blood units to medical facilities. The blood bank conveys the designated blood units to the distribution center, which thereafter distributes them to medical centers.

3.1. Mathematical Modeling

This concept aims to reduce blood shortages, promote health equity, limit wastage due to expiration, and optimize the allocation of new blood to fulfill specific demands. We integrate these aims by creating a detailed mathematical model. The development of the mathematical model follows the sequential phases outlined in Section 3. In Figure 2 a schematic illustration of the BSC used in developing the model discussed in this research. In this schematic elements of a typical blood supply chain network are shown by using standard charts symbols. This visual aid aims to improve the reader's understanding and offer a glimpse into the authors' thought process. To enhance the model's resemblance to real-world scenarios and manage some variables that cannot be accounted for in the mathematical model, certain assumptions are suggested and incorporated throughout the modeling process.

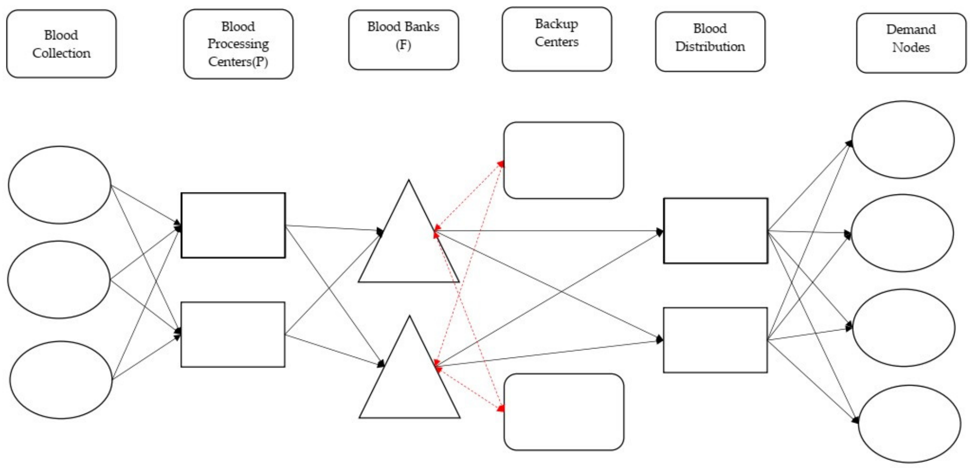

Figure 2. Typical blood supply chain network.

Assumptions

We propose and consistently use a set of assumptions during the modeling process to improve the model's relevance to real-world situations and handle factors that are difficult to include directly in the mathematical framework.

1. Daily, at regular predetermined intervals, the same equipped vehicles transport collected blood from collection centers to the blood processing facility.
2. Labor, machines, and equipment are available without limit.
3. Only the blood bank stores blood.
4. Blood collection centers, processing centers, and blood banks have limited capacity.
5. Under all circumstances, the blood transfer time from the blood collection centers to the blood processing center remains constant.
6. Blood shelf life is limited and calculated from donation time.
7. The time required to perform blood qualification tests is deterministic (1 day) and is deducted from the blood shelf life.

8. Blood in the first half-life is considered fresh; blood in the second half-life is considered ordinary.
9. The supplier receives the products in fresh form.
10. Medical experts determine the priority for blood demands.
11. The demand backorder in urgency classes 1 is not allowed, and each period demand must be satisfied in the same period.
12. The shortage of blood demand with urgency classes 1 (urgent due to disaster situation, urgent due to other reasons) is not allowed.
13. Assigned blood demands to hospitals are only sent from the blood distribution center, and lateral transshipment between hospitals is not allowed.
14. Initial inventory is negligible.
15. One bag of blood, or 450 cc, serves as the unit of measurement for blood quantity.
16. Assigned blood for each demand node is planned to be delivered in the distribution center.
17. We assume that all blood collection centers are active.
18. Capability for blood processing is greater than daily collecting capacity.
19. Reserved inventory is considered based on the planning cycle to address the needs of first-priority patients.

This section presents the notation used for mathematical modeling in both forecasting and optimization models. Each step defines the required notations to enhance clarity.

3.1.1. Forecasting Model

- Sets and Indices

1. A: Set of blood collection centers indexed by a, $A = \{1.2.\ldots.a.\ldots|A|\}$
2. H: Set of demand nodes (Hospitals) indexed by h, $\mathrm{H} = \{1.2.\ldots.\mathrm{h}.\ldots|\mathrm{H}|\}$
3. B: Set of blood groups indexed by b and b′, $\mathrm{B} = \{1.2.\ldots.\mathrm{b}.\ldots|\mathrm{B}|\}$
4. T: Planning period, $\mathrm{T} = \{1.2.\ldots.\mathrm{t}.\ldots|\mathrm{T}|\}$
5. U: Set of urgency classes, $\mathrm{U} = \{1.2.3\}$
6. I: Set of remaining blood shelf life (day), $\mathrm{I} = \{1.2.\ldots.\mathrm{i}.\ldots|\mathrm{I}|\}$

- Parameters

1. $\mathrm{AS}_{\mathrm{bait}}$: Actual Supply of blood group b in blood collection center a with remaining shelf life I at time t (Unit)
2. $\mathrm{AD}_{\mathrm{bhut}}$: Actual Demand for blood group b at hospital h with urgency class u in time t (Unit)

- Variables and Outputs

1. $\mathrm{ED}_{\mathrm{bhut}}$: Estimated Demand for blood group b at hospital h with urgency class u in time t (Unit)
2. $\mathrm{EDU}_{\mathrm{bhut}}$: Estimated Demand upper bound for blood group b at hospital h with urgency class u in time t (Unit)
3. $\mathrm{EDL}_{\mathrm{bhut}}$: Estimated Demand lower bound for blood group b at hospital h with urgency class u in time t (Unit)
4. $\mathrm{ES}_{\mathrm{bait}}$: Estimated Supply of blood group b in blood collection center a with remaining shelf life I at time t (Unit)
5. $\mathrm{ESU}_{\mathrm{biat}}$: Estimated Supply upper bound of blood group b in blood collection center a with remaining shelf life I at time t (Unit)
6. $\mathrm{ESL}_{\mathrm{biat}}$: Estimated Supply lower bound of blood group b in blood collection center a with remaining shelf life I at time t (Unit)

3.1.2. Scenario Selection Model

After selecting a proper forecasting model, we use it to forecast the supply and demand of blood. We compare these values using a predefined scenario tree and then select an appropriate scenario based on the results. Also, for deciding on transferring overcapacity collected blood to backup facilities or requesting blood from them to cover the shortage of blood, a hybrid method is used. In the hybrid method, if the forecast value for blood demand with priority 1, which, as previously mentioned, prohibits shortages, exceeds the forecast value for blood supply or if the actual demand exceeds the forecasted supply value, then permission to request blood from backup facilities is granted. Additionally, when the actual supply values surpass the total storage capacity, the transfer of surpluses to backup facilities becomes permissible. The optimization step determines the optimal value for transferring from/to backup facilities.

- Sets and Indices

1. F: Set of blood banks indexed by f, $F = \{1.2.\ldots.f.\ldots.|F|\}$
2. P: Set of blood processing centers indexed by p, $P = \{1.2.\ldots.p.\ldots|P|\}$
3. V: Set of backup facilities, $V = \{1.2.\ldots.v.\ldots|V|\}$
4. Sc: Set of possible scenarios indexed by s, $Sc = \{1.2.\ldots.s.\ldots|S|\}$

- Parameters

1. AS_{bift}: Actual Supply of blood group b in blood bank f with remaining shelf life I at time t (Unit)
2. Sf_{bpfit}: Quantity of blood with blood group b that is accepted during qualification tests and transferred from blood processing center p to blood bank f with remaining shelf life I at time t (Unit)
3. Variables and Outputs
4. In_{bift}: Inventory of blood group b with age (Shelf life) i in blood bank f at time t (Unit)
5. R_v: 1 if required to request blood from supporting facility v, otherwise 0
6. S_{fv}: 1 if required to send blood from Blood bank f to supporting facility v, otherwise 0

To account for all possible conditions, we consider a scenario selection flow chart (Figure 3), which takes into account existing conditions and compares forecast and actual values. We will take actions based on each scenario to ensure the sustainability of BSC. In this step, the actual supply is calculated by adding up the inventory quantity from the previous day and the blood that has been transferred to the blood bank today.

Actual Supply:

$$\sum_{b\in B}\sum_{f\in F} AS_{bift} = \sum_{b\in B}\sum_{f\in F} In_{bif(t-1)} + \sum_{b\in B}\sum_{p\in P}\sum_{f\in F} Sf_{bpfit}\,;\ \forall\ \ i \in I-\{1\}.\ t\in T-\{1\} \quad (1)$$

Scenarios are established using forecasts and actual data integrated into the decision support system. We conduct demand analysis in the first step to compare the actual demand to the upper and lower bounds of the forecasted demand. Subsequently, the appropriated demand status will result in a similar analysis for the supply of blood, utilizing both forecasted and actual values calculated through Equation (1). The outcome of the previously mentioned step will identify the relevant scenario and necessary actions anticipated to maintain BSC performance and enhance its resilience. Table 3 presents the scenarios, required actions, and values of R_v and S_{fv}, which are parameters of the model concerning supporting facilities and suppliers. Scenario characteristics may appear similar; however, they differ significantly, as each necessitates actions based on variations in inventory levels or discrepancies between forecasted and actual demand and supply values. A transfusion expert should supervise these actions and use the decision support system this paper proposes.

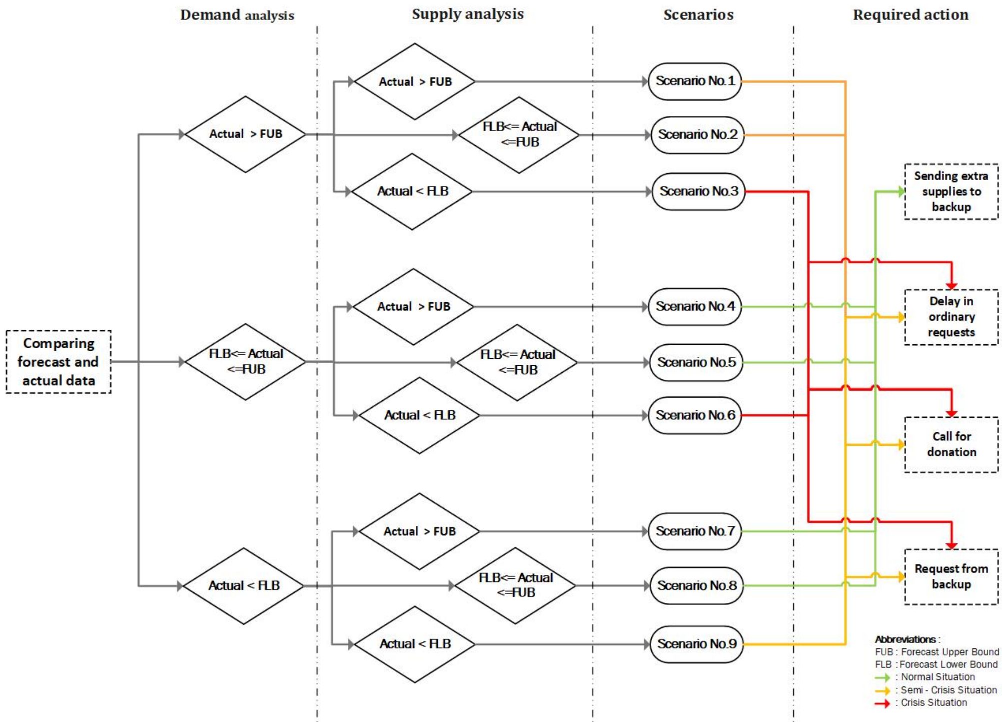

Figure 3. Scenario selection flow chart.

Table 3. Scenarios characteristics.

Scenario	Situation	Required Action(s)	R_v	S_{fv}
No. 1	Semi-Crisis	Delay in ordinary requests Call for donation Request from backup	1	0
No. 2	Semi-Crisis		1	0
No. 3	Crisis		1	0
No. 4	Normal	Sending extra supplies to backup	0	1
No. 5	Normal		0	1
No. 6	Crisis	Delay in ordinary requests Call for donation Request from backup	1	0
No. 7	Normal	Sending extra supplies to backup	0	1
No. 8	Normal		0	1
No. 9	Semi-Crisis	Delay in ordinary requests Call for donation Request from backup	1	0

3.1.3. Optimization Model

- Sets and Indices

1. G: Set of blood distribution centers indexed by g, $G = \{1.2.\ldots.g.\ldots|G|\}$
2. B': Set of substitutable blood groups b that can be substituted by b′, $S_{bb'} \subseteq B$.
3. B_f : Set of days for considering blood as Fresh; $B_f = \left\{1.2.\ldots\left|\frac{I}{2}\right|\right\}$
4. B_{or} : Set of days for considering blood as Ordinary; $B_{or} = \left\{\left|\frac{I}{2}\right|.\ldots|I|\right\}$

- Parameters

1. Cw_f: Blood bank f total capacity (Unit)
2. Cv_v: backup facility v total capacity (Unit)
3. $WP_{bb'}$: Substitution priority for substituting blood group b with blood group b'

4. RI_{bift}: Reserved inventory for blood group b in blood bank f with remaining shelf life I at time t
5. Dv_{bvut}: Supporting Facility v actual demand for blood group b with priority u at time t

- Variables and Outputs

1. Sg_{bfgit}: Quantity of blood with blood group b that is transferred from blood bank f to blood distribution center g with remaining shelf life I at time t (Unit)
2. Sh_{bhiut}: Quantity of blood with blood group b with age i that is assigned to hospital h to fulfill demand with urgency class u at time t (Unit)
3. Od_{bft}: Quantity of blood with blood group b that is outdated in blood bank f at time t (Unit)
4. Ud_{bhut}: Total unfulfilled demand in hospital h of blood group b with urgency class u at time t (Unit)
5. $Sh'_{b'hiut}$: Quantity of blood with blood group b' with remaining shelf life I that is substituted with blood group b to fulfill hospital h demand with urgency class u at time t (Unit)
6. Se_{bvit}: amount of blood with blood group b with remaining shelf life i at time t is assigned to supporting facilities v to fulfill the demand (Unit)
7. Sv_{bifvt}: amount of blood with blood group b with remaining life span i to be sent from blood bank f to supporting facilities v at time t (Unit)
8. Rv_{bvt}: Quantity of blood with blood group b requested from backup supplier v at time t (Unit)
9. $W_{bb'}$: 1 if required to substitute blood group b with blood group b', otherwise 0

3.1.4. The Objective Functions

Given that blood is a perishable commodity with no viable alternatives and considering that blood transfusion institutions in countries such as Iran operate on a non-profit basis, the objective functions employed in this study are non-monetary in nature. In addition, the optimization of non-monetary objective functions will also result in the optimization of associated expenses.

1. Minimum shortage

We set this objective function to minimize blood shortage when responding to requests from demand nodes. As mentioned earlier, shortage is not allowed for first urgency class demands.

$$Min\ Z_S = \sum_{b\in B}\sum_{h\in H}\sum_{\substack{u\in U\\ .\ u\neq 1}}\sum_{t\in T} Ud_{bhut} \tag{2}$$

2. Minimum outdate

The second object function focuses on reducing blood waste from out-of-date and expired blood in blood banks.

$$Min\ Z_o = \sum_{b\in B}\sum_{f\in F}\sum_{t\in T} Od_{bft} \tag{3}$$

3. Maximum freshness

The last objective function is to minimize the utilization of non-fresh blood by introducing a coefficient that accounts for the utilization of fresh blood. Consequently, the implementation prioritizes maximizing the utilization of fresh blood. We use the coefficient $\frac{i-\frac{|I|}{2}}{|I|}$ to quantify the impact of using fresh blood in response to requests.

$$Min\ Z_f = \sum_{b\in B}\sum_{h\in H}\sum_{i\in I}\sum_{u\geq 2}\sum_{t\in T}\frac{i-\frac{|I|}{2}}{|I|} * Sh_{bhiut} + \sum_{b'\in B'}\sum_{h\in H}\sum_{i\in I}\sum_{u\geq 2}\sum_{t\in T}\frac{i-\frac{|I|}{2}}{|I|} * Sh'_{b'hiut} \tag{4}$$

3.1.5. Model Constraints

$$\sum_{b\in B}\sum_{p\in P}\sum_{i\in I}Sf_{bpfit}+\sum_{b\in B}\sum_{i\in I}\sum_{v\in V}Sv_{bifvt}+\sum_{b\in B}\sum_{i\in I}In_{bif(t-1)} \geq \sum_{b\in B}\sum_{i\in I}\sum_{g\in G}Sg_{bfgit}+\sum_{b\in B}Od_{bft}+\sum_{b\in B}\sum_{v\in V}\sum_{i\in I}Se_{bvit};\forall\ f\in F.\ t\in T \quad (5)$$

$$\sum_{b\in B}\sum_{p\in P}\sum_{i\in I}Sf_{bpfit}+\sum_{b\in B}\sum_{i\in I}\sum_{v\in V}Sv_{bifvt}-\sum_{b\in B}\sum_{i\in I}In_{bif(t-1)}-\sum_{b\in B}Od_{bft}\ \leq\ Cw_f\ ;\ \forall\ f\in F.\ t\in T \quad (6)$$

$$\sum_{b\in B}\sum_{i\in I}\sum_{f\in F}Sv_{bifvt}*S_{fvt}+\sum_{b\in B}\sum_{i\in I}Se_{bvit}\leq Cv_v\ ;\forall\ v\in V.\ t\in T \quad (7)$$

$$\sum_{b\in B}\sum_{i\ \in Bf}\sum_{f\in F}Sv_{bifvt}*S_{fvt}\leq\sum_{b\in B}Rv_{bvt}*R_{vt}\ ;\ \forall\ v\in V.\ t\in T \quad (8)$$

$$\sum_{b\in B}\sum_{i\ \in Bf}Se_{bvit}\ \leq\ \sum_{b\in B}\sum_{u\ \geq 2}Dv_{bvut}\ ;\ \forall\ v\in V.t\in T \quad (9)$$

$$\sum_{b\in B}\sum_{f\in F}\sum_{g\in G}\sum_{i\in I}Sg_{bfgit}=\sum_{b\in B}\sum_{h\in H}\sum_{i\in I}\sum_{u\in U}Sh_{bhiut}+\sum_{b'\in\ B'}\sum_{h\in H}\sum_{i\in I}\sum_{u\in U}Sh'_{\ b'hiut};\forall\ t\ \in T \quad (10)$$

$$\sum_{b\in B}\sum_{h\in H}\sum_{u\in U}AD_{bhut}=\sum_{b\in B}\sum_{h\in H}\sum_{i\in I}\sum_{u\in U}Sh_{bhiut}+\sum_{b'\in\ B'}\sum_{h\in H}\sum_{i\in I}\sum_{u\in U}Sh'_{\ b'hiut}+\sum_{b\in B}\sum_{h\in H}\sum_{u\in U}Ud_{bhut}\ ;\ \forall\ t\ \in T \quad (11)$$

$$\begin{aligned}\sum_{b\in B}\sum_{i\in I}\sum_{f\in f}In_{bift}=&\sum_{b\in B}\sum_{i\in I}\sum_{f\in f}In_{bif(t-1)}+\sum_{b\in B}\sum_{p\in P}\sum_{f\in F}\sum_{i\in I}Sf_{bpfit}\\ &-\Bigg(\sum_{b\in B}\sum_{f\in F}Od_{bft}+\sum_{b\in B}\sum_{h\in H}\sum_{i\in I}\sum_{u\in u}Sh_{bhiut}+\sum_{b'\in\ B'}\sum_{h\in H}\sum_{i\in I}\sum_{u\in u}Sh'_{\ b'hiut}+\sum_{b\in B}\sum_{v\in V}\sum_{u\in U}AD_{bvut}\\ &+\sum_{b\in B}\sum_{i\in i}\sum_{f\in F}\sum_{v\in V}Sv_{bifvt}+\sum_{b\in B}\sum_{v\in V}\sum_{i\in I}Se_{bvit}\Bigg);\ \forall\ t\ \in T\end{aligned} \quad (12)$$

$$\sum_{b\in B}\sum_{i\in I}\sum_{f\in f}In_{bift}\ \leq\ \sum_{f\in F}Cw_f\ ;\ \forall\ t\ \in T \quad (13)$$

$$\sum_{b\in B}\sum_{i\in I}\sum_{f\in f}In_{bift}\ \geq\sum_{b\in B}RI_{but};\ \ \forall\ t\ \in T \quad (14)$$

$$\sum_{b\in B}\sum_{f\in F}Od_{bft}=\sum_{b\in B}\sum_{f\in F}In_{bif(t-1)};\ i\in\{|I|\}\ .\ t\in T-\{1\} \quad (15)$$

$$\sum_{b\in B}\sum_{h\in H}\sum_{u\in\{1\}}Ud_{bhut}>0\Longrightarrow\sum_{b\in B}\sum_{h\in H}Sh_{bhiut}=0\ ;\ \forall\ i\ Bor\ .t\in T.\ u\in\{1\} \quad (16)$$

$$\sum_{b\in B}\sum_{h\in H}\sum_{u\in\{2\}}Ud_{bhut}>0\ \Rightarrow\ \sum_{b\in B}\sum_{f\in F}\sum_{i\ \in Bf}In_{bift}>0\ \Longrightarrow\ \sum_{b\in B}\sum_{h\in H}Sh_{bhiut}=0\ ;\ \forall\ i\ \in Bor\ .t\in T.\ u\in\{2\} \quad (17)$$

$$\sum_{b\in B}\sum_{h\in H}\sum_{u\in\{2\}}Ud_{bhut}>0\ \Rightarrow\ \sum_{b\in B}\sum_{f\in F}\sum_{i\ \in Bf}In_{bift}=0\ \Longrightarrow\ \sum_{b\in B}\sum_{h\in H}Sh_{bhiut}=0\quad ;\ \forall\ i\ \in Bf\ .t\in T.\ u\in\{2\} \quad (18)$$

$$\sum_{b\in B}\sum_{h\in H}\sum_{u\in\{3\}}Ud_{bhut}>0\ \Rightarrow\ \sum_{b\in B}\sum_{f\in F}\sum_{i\ \in Bor}In_{bift}>0\ \Longrightarrow\ \sum_{b\in B}\sum_{h\in H}Sh_{bhiut}=0\ ;\ \forall\ i\ \in Bf\ .t\in T.\ u\in\{3\} \quad (19)$$

$$\sum_{b\in B}\sum_{h\in H}\sum_{u\in\{3\}}Ud_{bhut}>0\ \Rightarrow\ \sum_{b\in B}\sum_{f\in F}\sum_{i\ \in Bor}In_{bift}=0\ \Longrightarrow\ \sum_{b\in B}\sum_{h\in H}Sh_{bhiut}=0\ ;\ \forall\ i\ \in Bf\ .t\in T.\ u\in\{3\} \quad (20)$$

$$\sum_{b\in B}\sum_{h\in H}\sum_{u\in\{1\}}Ud_{bhut}>0\Rightarrow\sum_{b'\in\ B'}\sum_{h\in H}Sh'_{\ b'hiut}=0\ ;\forall\ i\in I.\ u\in\{1\}\ .\ t\in T \quad (21)$$

$$\sum_{b\in B}\sum_{h\in H}\sum_{u\in\{2\}} Ud_{bhut} > 0 \Rightarrow \sum_{b\in B}\sum_{h\in H} Sh_{bhiut} + \sum_{b'\in\ B'}\sum_{h\in H} Sh'\ _{b'hiut} = 0\ ;\forall\ i\in I\ .\ u\in\{3\}.t\in T \tag{22}$$

$$\sum_{b'\in B}\sum_{h\in H}\sum_{i\in I} Sh'\ _{b'hiut} = \sum_{b\in B}\sum_{b'\in B}\sum_{h\in H} W_{b\ b'} *\ Wp_{bb'} * Ud_{bhut};\ \forall\ u\in\{2.3\}\ .\ t\in T \tag{23}$$

$$Sg_{bfgit}.\ Sh_{bhiut}.\ In_{bift}.\ Od_{bft}.\ Ud_{but}.Sh'\ _{b'hiut}.\ Se_{bvit}.\ Sv_{bifvt}.\ Rv_{bvt}.\ \geq 0 \tag{24}$$

$$S_{fv}\ .\ W_{b\ b'}.\ R_v\ \in\ \{0\ .\ 1\} \tag{25}$$

Equations (5) and (6) specify that the quantity of blood units transferred from the processing center to each blood bank will not exceed the capacity of that blood bank, considering its current inventory. In these constraints, the input quantity of blood comes from the blood processing centers and support centers (as requested), while the output is the blood leaving each blood bank, taking into account the presence of a reserve inventory for emergency conditions.

The model incorporates Equations (7)–(9) to ensure that support centers mobilize to address crisis situations in the blood supply chain, such as an increase in demand or a shortage. We design these constraints to exclusively utilize support centers in emergency situations. Additionally, these support centers duly fulfill their requests in crisis conditions. Given that the delivery of blood units to healthcare facilities occurs at distribution centers (blood distribution units), the input quantity to these centers is equal to the allocated amount to healthcare institutions, including compatible and homogeneous blood groups. This constraint guarantees the exclusive availability of blood to healthcare facilities via this route, preventing any transfer between them.

We have added Equation (10) to the model to enforce this condition. Equation (11) guarantees that the centers fulfill their requests according to the established policies, with a maximum limit that matches the requested quantity. Equations (12)–(14) calculate the daily blood inventory across all blood banks, ensuring the presence of a reserve inventory and preventing the occurrence of blood exceeding capacity. Equation (15) calculates the quantity of expired blood. We consider blood that reaches the end of its lifespan on the previous day to have expired on the following day, and we must deduct it from the daily blood inventory.

Clinicians allocate blood based on defined priorities, considering the emergency status and patient needs. We must first attend to all requests with the highest priority and then proceed to process and complete those with lower priority levels. The feasibility of processing requests from lower priorities is contingent upon the complete satisfaction of the requirements associated with the highest priority. We must emphasize that the most urgent demands cannot tolerate shortages. We have included Equations (16)–(20) to ensure the model aligns with the specified allocation policy. Equations (21)–(23) perform blood substitution based on medical preferences and establish prioritization based on the patients' conditions. The highest priority does not allow blood substitution, but the second and third priorities do.

Equations (24) and (25) serve as the constraint for enforcing all variables to be integer and non-negative.

3.2. Solution Approach

We have developed a multi-objective non-linear mixed-integer optimization (MONLMIP) model for applicability in diverse settings. We have designed this model to function effectively in normal, semi-crisis, or crisis circumstances, which arise from the first and second steps of the suggested method. Consequently, it should serve as a model not only for operating in such circumstances but also for managing excess or deficiency of inventory units to uphold BSC performance and ensure patient health in accordance with the needs of the patient. To make sure the model can be used in a variety of situations and meets different needs, as suggested by system experts during scenario selection, it is important to use a solution procedure that is adaptable and can handle changes in model settings.

Concerning the authors, it is considered suitable to solve the model by converting the MONLMIP into a single-objective NLMIP and assigning weights to each objective function. This method is a common approach for solving multi-objective problems, characterized by its simplicity, adaptability in prioritizing objectives, and compatibility with existing optimization solvers. Assigning weights transforms the multi-objective framework into a scalar optimization problem, simplifying its implementation and enabling the use of standard optimization techniques.

The model is classified as a deterministic Mixed Integer Programming (MIP) by establishing accordance weights and utilizing the big M approach, consistent with prevalent deterministic models in the literature [2]. The model addresses a real-world problem characterized by various factors and variables classed as NP-Hard. Metaheuristic approaches are especially effective for addressing large-scale optimization problems due to their ability to efficiently explore vast search spaces.

We apply the Genetic Algorithm (GA), a population-centric metaheuristic algorithm built on the principles of natural selection. Genetic algorithms produce consecutive iterations of solutions by combining and altering them to explore the solution space. Its ability to equilibrate exploration and exploitation makes it very proficient in identifying high-quality answers in complex optimization problems such as the one studied in this article. The procedure involves establishing a fitness function, implementing penalties, and producing modified fitness values to direct the search process.

We employ Particle Swarm Optimization (PSO) as a benchmark to validate the proposed solution approach. Particle Swarm Optimization (PSO) is an optimization method derived from the social dynamics of swarms. It conceptualizes the issue as a search space wherein particles progressively modify their placements and velocities based on personal experience and that of their peers. PSO is renowned for its simplicity, swift convergence, and efficacy in tackling non-linear problems, rendering it an optimal comparative framework for assessing GA's performance.

Both genetic algorithms (GA) and particle swarm optimization (PSO) have shown to be effective in addressing issues pertaining to the optimization and simulation of the blood supply chain (BSC) in the available literature [2]. Their adaptability and shown performance in analogous applications highlight their appropriateness for the proposed model, guaranteeing a balance between computational efficiency and solution quality.

The proposed solution methodology proceeds as follows:

1. Weight Assignment for Objective Functions: The MONLMIP model is converted into a single-objective NLMIP model through the assignment of weights to each objective function. This step simplifies the multi-objective framework to a scalar optimization problem, facilitating handling and alignment with standard optimization methods.
2. Linearization, The Big M Method is employed to tackle the non-linearity present in the model. This transformation reformulates the model into a linear format, consistent with established methods for addressing deterministic models.
3. Application of Genetic Algorithms: The Genetic Algorithm is utilized to address the linearized model. This involves establishing the fitness function, incorporating penalties for constraint violations, and calculating adjusted fitness values for guiding the search for optimal solutions. GA's strong exploration and exploitation capabilities render it highly effective for large-scale, non-linear optimization problems.
4. Validation through Comparison with PSO: Additional experiments are conducted to evaluate the robustness and efficacy of the GA-based solution using PSO as a benchmark. The strengths of PSO, including its simplicity, rapid convergence, and established effectiveness in addressing similar optimization challenges, render it an appropriate benchmark for assessing the performance of GA.

- Defining Weights

The variable W_i is defined to represent the weight of each objective function. This strategy involves the involvement of system experts' judgments in any given scenario. We conducted multiple meetings with specialists from the IBTO transfusion and hemovigilance departments to determine the weights for the linearization phase and to identify acceptable trade-offs between objectives and functions. These specialists provided valuable insights based on their practical and medical knowledge in managing the blood supply system.

The Analytic Hierarchy Process (AHP) is employed to evaluate the respective importance of each objective function in different predetermined scenarios. The input for the AHP was collected from a panel of experts with extensive experience and knowledge in laboratory hematology, blood supply chain, and healthcare logistics. Twelve professionals with significant expertise in blood transfusion operations and logistics contributed to this evaluation, including administrators of the Central Blood Bank, operational practitioners, the field operations manager, and the research department of IBTO. The variety of skills guaranteed a thorough comprehension of the requirements and their significance in both regular and emergency situations. Each expert offered pairwise comparisons derived from their practical and academic expertise, enhancing a comprehensive and equitable weighing system.

We used pairwise comparisons to systematically assess the relevance of each objective function, ensuring consistency in expert evaluations. Table 4 presents the weights that the AHP technique achieved consensus on. We conducted sensitivity analyses to assess the robustness of these weights, demonstrating that minor changes did not significantly affect the model's results. This methodology enhances the stability and reliability of outcomes by aligning scenarios with practical challenges in the blood supply chain.

Table 4. Scenarios objective functions' weights.

Scenario	W_1	W_2	W_3
No. 1	0.6	0.3	0.1
No. 2	0.6	0.3	0.1
No. 3	1	0	0
No. 4	1/3	1/3	1/3
No. 5	1/3	1/3	1/3
No. 6	1	0	0
No. 7	1/3	1/3	1/3
No. 8	1/3	1/3	1/3
No. 9	0.6	0.3	0.1

- Linearization: The presence of Equation (23) in the model leads to a non-linear function. The big M approach is employed to convert Equation (23) into a linear one, thus decreasing both the time required to solve the model and its complexity.

$$\begin{gathered} y_{b\ b'} = W_{b\ b'} * Wp_{bb'} \\ y_{b\ b'} \leq W_{b\ b'} \\ y_{b\ b'} \leq Wp_{bb'} \\ y_{b\ b'} \geq W_{b\ b'} + Wp_{bb'} - M * (1 - W_{b\ b'}) \end{gathered} \tag{26}$$

Equation (27) is substituted with Equation (23) by incorporating the aforementioned limitation and substituting $y_{bb'}$.

$$\sum_{b' \in B} \sum_{h \in H} Sh'_{b'hiut} = \sum_{b' \in B} \sum_{h \in H} y_{bb'} * Ud_{bhut}; \forall u \in \{2.3\}.t \in T \tag{27}$$

- Fitness Function, Penalty, and Adjusted Fitness

The fitness function is common between both GA and PSO, representing the aggregated objective to be optimized. It is defined as follows:

$$Fitness(x) = W_1 * \sum_{b\in B}\sum_{h\in H}\sum_{\substack{u\in U\\ .\ u\neq 1}}\sum_{t\in T} Ud_{bhut} + W_2 * \sum_{b\in B}\sum_{f\in F}\sum_{t\in T} Od_{bft} + W_3 * \sum_{b\in B}\sum_{f\in F}\sum_{t\in T} Od_{bft} \quad (28)$$

To handle constraints, a penalty mechanism is applied. This assigns a penalty score to infeasible solutions based on the magnitude of constraint violations:

$$Penalty = \sum_{j=1}^{m} max\left(0.Violation_j\right) \quad (29)$$

where $Violation_j$ quantifies the extent of violation for each constraint.

The final fitness value, accounting for penalties, is given by:

$$Adjusted\ Fitness = Fitness - Penalty \quad (30)$$

This ensures that feasible solutions are favored over infeasible ones, guiding the optimization process toward valid regions of the search space.

- Genetic Algorithm (GA)

The genetic algorithm is employed to solve the single objective linear MIP problem. The GA is well suited for this type of problem due to its ability to efficiently explore large and complex solution spaces and provide near-optimal solutions for conflicting objectives. Additionally, its inherent stochastic nature makes it capable of escaping local optima, which is a common challenge in non-linear problems.

Encoding and Initialization

The problem variables are encoded into chromosomes, where each chromosome represents a potential solution. An initial population is generated randomly or based on heuristic methods to ensure diversity in the search space. Proper initialization is critical to improving convergence rates and avoiding premature stagnation.

GA Parameters: Key GA parameters include the population size, crossover rate, mutation rate, and the number of generations. These parameters are fine-tuned to balance exploration and exploitation, ensuring that the algorithm converges to high-quality solutions within a reasonable computational time.

- Particle Swarm Optimization (PSO)

PSO is inspired by the social behavior of swarms, where particles move through the search space to find the best solution. Each particle adjusts its position based on its own experience (p_{best}) and the swarm's collective experience (g_{best}). PSO procedure for solving weighted single-objective problems is as follows:

1. Initialization:

   Randomly initialize particle positions $x_{i.0}$ and velocities $v_{i.0}$.
   Set initial p_{best} for each particle and identify the global best g_{best}.

2. Fitness Evaluation:

   Calculate the adjusted fitness for each particle using the weighted fitness function.

3. Update Personal Best:

   If a particle's fitness is better than its p_{best}, update p_{best}.

4. Update Global Best:

   Update g_{best} as the best position among all particles.

5. Update Velocities and Positions:

   Adjust particle velocities using the following equations:

$$v_{i.k+1} = wv_{i.k} + c_1 r_1 (p_{best} - x_{i.k}) + c_2 r_2 (g_{best} - x_{i.k}) \tag{31}$$

where w is the inertia weight, and c_1 and c_2 are cognitive and social coefficients.

Update positions:

$$x_{i.k+1} = x_{i.k} + v_{i.k+1} \tag{32}$$

6. Termination:

   Repeat until reaching the maximum number of iterations or no improvement in g_{best}.

3.3. Results

Until the proposed model, which prioritizes the need for blood, is actually put into practice, the results of the model are only examined to meet each objective function. A sample dataset is constructed to validate the model, which includes data relating to collection and demand centers for a one-year period. The integrated problem is being tested using the provided sample dataset. The results indicate that the proposed model can accurately forecast the future status of the model using historical data. Additionally, it successfully captures the model status throughout the scenario-selection phase in order to determine the weights of each objective function. The findings for each scenario demonstrated the successful implementation of the authors' initial proposal to prioritize the demand and allocate blood based on a policy of prioritizing classes and substituting blood groups.

We performed the calculations on a laptop with an Intel Core i7 4710HQ CPU and 8 GB of RAM. The software used includes Python for optimization and time-series packages and MySQL 8.0 for the data warehouse. The principal packages include Pyomo for mathematical formulation and optimization, Statsmodels for time-series modeling, and Matplotlib and Seaborn for visualization.

Case Study

In order to validate the practicality of the proposed model, a case study is presented in this section. The Iranian Blood Transfusion Organization (IBTO), established in 1975, is the only authorized source of blood in the country. IBTO has subsidiaries in each province. The Tehran Province branch of IBTO is responsible for supplying blood to more than 150 hospitals through 12 collection centers (Figures 4 and 5). Blood processing and storage are centralized in the VESAL centers. All hospital requests are processed and handed over to hospitals at the blood distribution centers, which are also located in the VESAL Center. In this research, the performance of the proposed model is evaluated using data from the Tehran branch of IBTO, which serves more than 150 hospitals and medical centers distributed in the Tehran Province.

During the starting point of the model, the utilization of ARIMA and SARIMA for forecasting the quantity of blood donated at donation centers and demand nodes (hospitals) revealed that the time series will be influenced by factors such as national holidays, moral considerations, and even weather conditions. Based on these criteria, the seasonal form of the ARIMA model (SARIMA) is more effective than the non-seasonal form in capturing the trends in both donations and demand. Additionally, SARIMA has the capability to make forecasts with lower error rates, as indicated in Table 5.

Figure 4. Tehran blood collection centers.

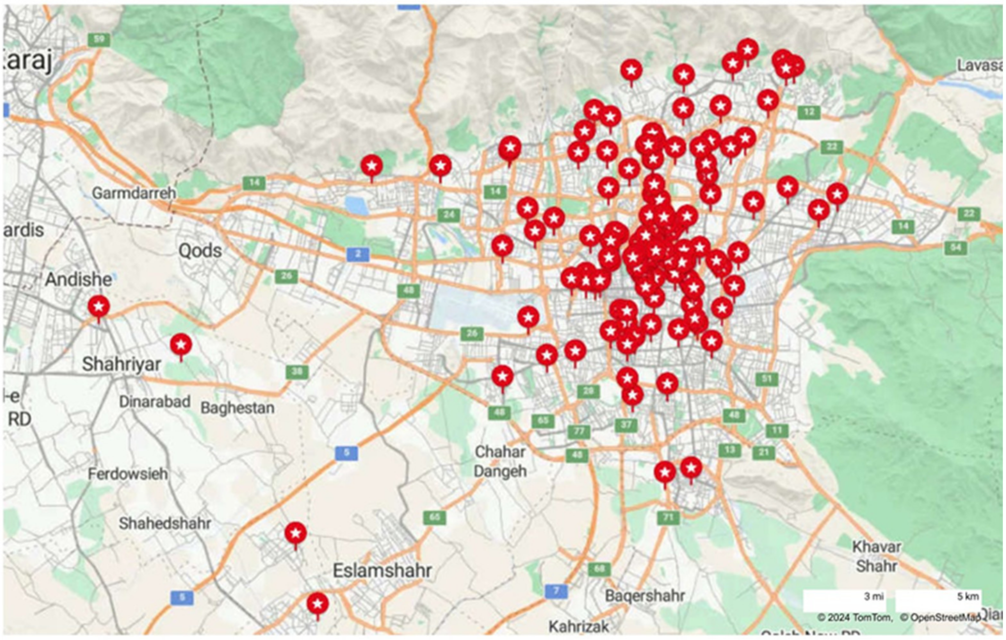

Figure 5. Tehran hospitals.

Table 5. Errors of ARIMA and SARIMA models.

Blood Group	ARIMA					SARIMA				
	O	A	B	AB	Total	O	A	B	AB	Total
MAE	44.82	41.65	20.2	10.23	101.74	37.87	43.86	17.07	8.64	76.63
RMSE	63.39	59.88	24.78	13.94	142.62	58.95	55.69	23.05	15.86	132.6

The review of the data reveals that the quantity of blood requested over time has been consistently conforming to a normal distribution pattern. From the first quarter of 2018 onward, there has been a rise in the need for blood units as a result of the COVID-19 virus outbreak and the subsequent pandemic. Additionally, it is disclosed that the demand for different blood groups varies. Blood group O has the highest frequency of demand among all blood groups, followed by blood groups A, B, and AB (Figures 6 and 7).

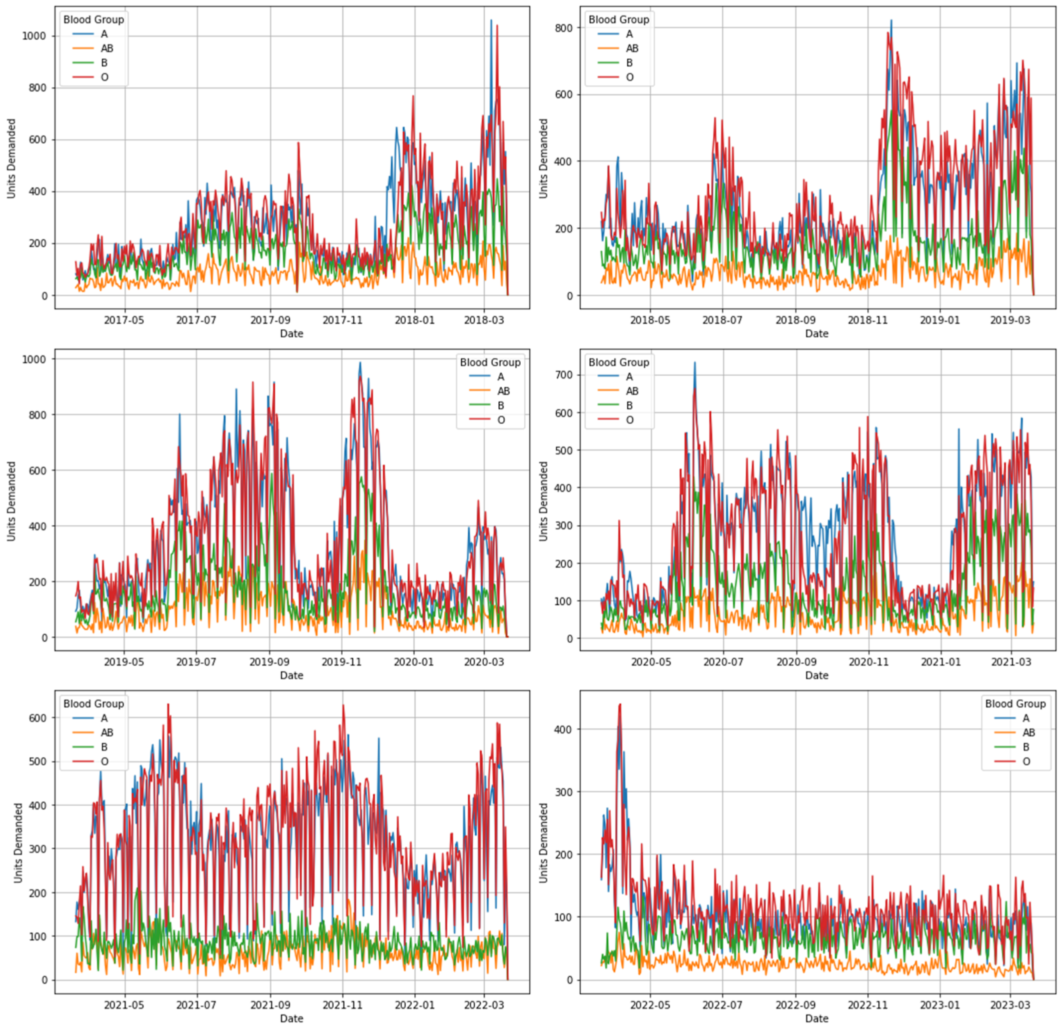

Figure 6. Blood units demanded per year for blood groups.

The forecast has been conducted using the 80/20 law. Initially, a subset of the dataset is reserved for testing the model in the future. The remaining dataset follows the 80/20 rule for forecasting. Out of the remaining dataset, 80% is used for training the model, and the remaining 20% shall be used for validation. The training data for the model ranged from March 2017 to January 2021, while the validation data encompassed the period from January 2021 to January 2022. Subsequently, the model's performance was validated; it is fitted to the entire dataset and used for forecasting the future.

After it was determined that the SARIMA model is the most suitable model for forecasting, the scenario-selection step was initiated. The simulation of each scenario was conducted using a set of data. As mentioned before, after determining each day's status by using the scenario tree weight of each objective function in order to make the multi-objective model into a single-objective one, it is set. Following this, with the solution approach discussed in Section 3.2, the model was solved. The result of solving the model in each scenario is presented in Table 6. To evaluate the GA performance and its robustness in multiple scenarios, we have used the partial swarm optimization (PSO) algorithm as the baseline. The results (Tables 6 and 7) revealed that GA consistently outperformed the baseline in achieving a higher overall objective function score. Figure 8 shows how the two methods resulted in various conditions and scenarios.

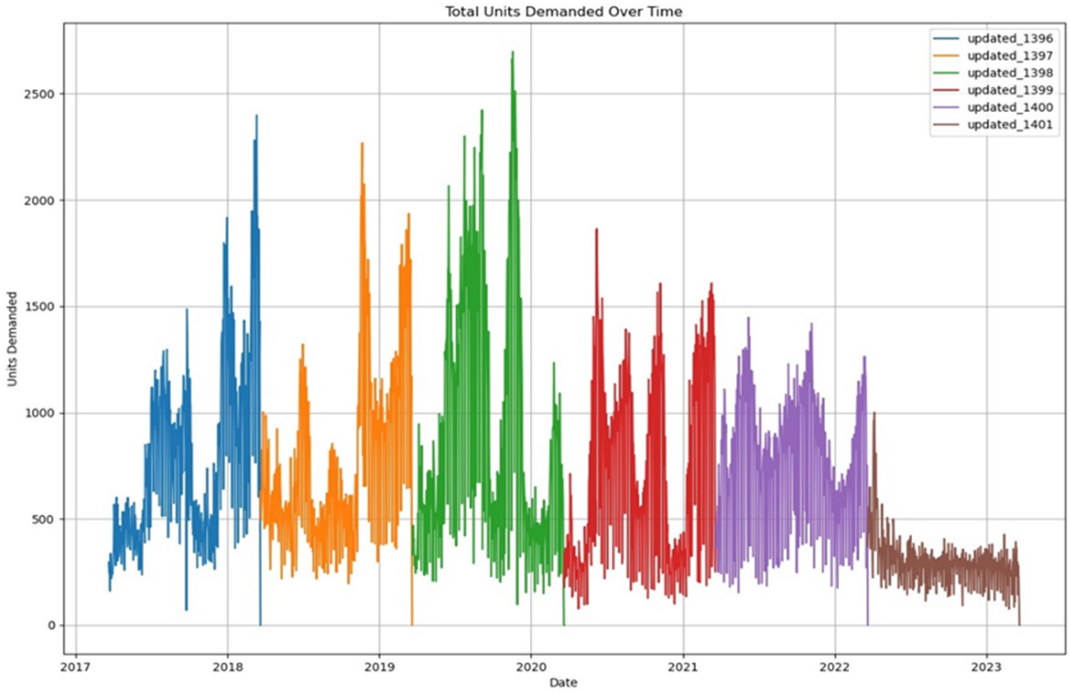

Figure 7. Total units demanded over time.

Table 6. GA algorithm's objective function results.

Scenario	Objective 1 Minimum Shortage	Objective 2 Minimum Outdate	Objective 3 Maximum Freshness	Combined Objective	Fitness Function
No. 1	240	630	567	1437	390
No. 2	1171	927	961	3059	1077
No. 3	816	1193	882	2891	816
No. 4	1095	539	263	1897	632
No. 5	369	421	1029	1819	606
No. 6	546	1182	404	2132	546
No. 7	330	757	541	1628	543
No. 8	697	402	1049	2148	716
No. 9	1072	827	624	2523	954

Table 7. PSO algorithm's objective function results.

Scenario	Objective 1 Minimum Shortage	Objective 2 Minimum Outdate	Objective 3 Maximum Freshness	Combined Objective	Fitness Function
No. 1	251	635	565	1451	398
No. 2	1175	933	955	3063	1080
No. 3	820	1195	880	2895	820
No. 4	1100	545	265	1910	637
No. 5	375	425	1025	1825	608
No. 6	550	1195	405	2150	550
No. 7	335	765	540	1640	547
No. 8	705	410	1050	2165	722
No. 9	1075	830	620	2525	956

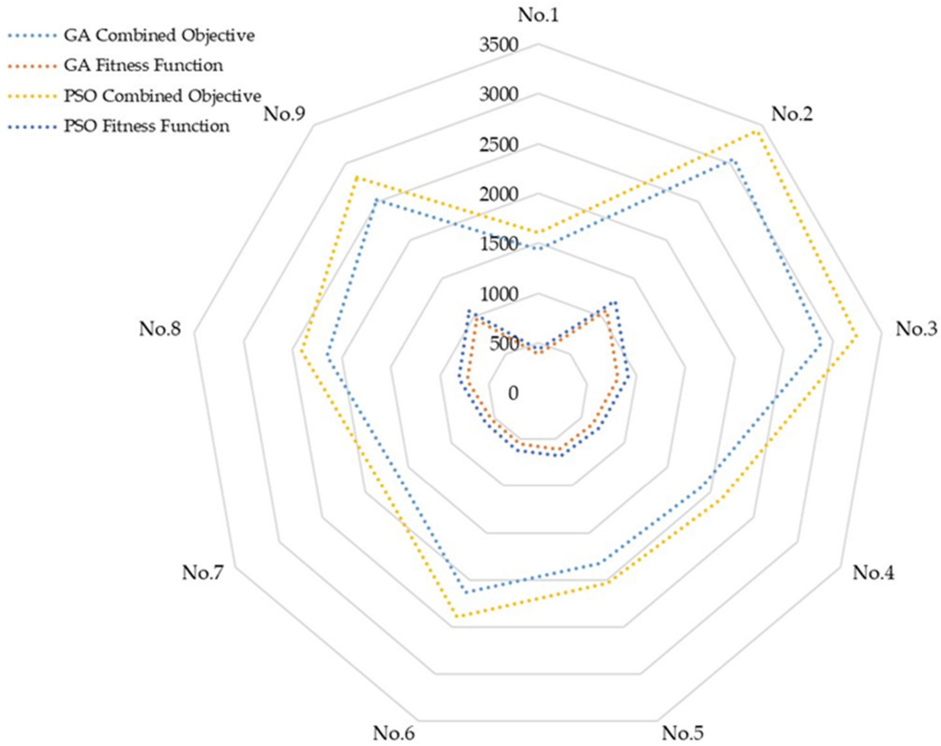

Figure 8. Comparison of fittness function values of GA and PSO.

4. Discussion

The forecasting method errors presented in Table 5 indicate that the SARIMA model is sufficiently effective in forecasting future blood supply and demand. The SARIMA model effectively captures the seasonal characteristics of the data and facilitates accurate forecasting. In the literature on blood forecasting, the focus is primarily on predicting either demand or supply, and there is a lack of integration of forecasting values into an optimization model.

No existing approach in the literature has proposed a framework that analyzes the current status of the BSC to provide system experts with a perspective. The absence of this model has resulted in a division between models designed for normal conditions and those intended for crisis situations. In this study, we analyze the BSC using a scenario selection flowchart (Figure 3), proposing managerial insights and recommendations for each identified scenario. This enables decision-makers to gain a clearer understanding of the system and to communicate necessary actions through the proposed framework.

Our study's mathematical model, which incorporates multiple levels of BSC, has resulted in a MONLMIP. Nearly all nonlinear models in the literature undergo conversion to linear ones. We apply the big-M approach similarly, enabling us to reduce complexity and avoid local optima. The weighted sum method has transformed the multi-objective model to include the second stage of the proposed study. This method enables the model to capture the characteristics of each scenario, resulting in it being resilient to real-world changes and enhancing its applicability. We establish the compliance fitness function and the associated penalty. The PSO algorithm serves as the baseline optimization model for a comparison with the GA algorithm, which is the primary solution approach. The GA method may identify near-optimal solutions for complex problems with numerous variables and parameters in a logical way. Table 6 and Figure 9 present the outcomes of the GA algorithm implementation. Compared to the PSO method results presented in Figure 10, it has been demonstrated that the GA method achieves better results. Figure 8 presents a comparative depiction of the fitness functions for both PSO and GA. The genetic algorithm evidently outperforms the particle swarm optimization in several instances while addressing our proposed model.

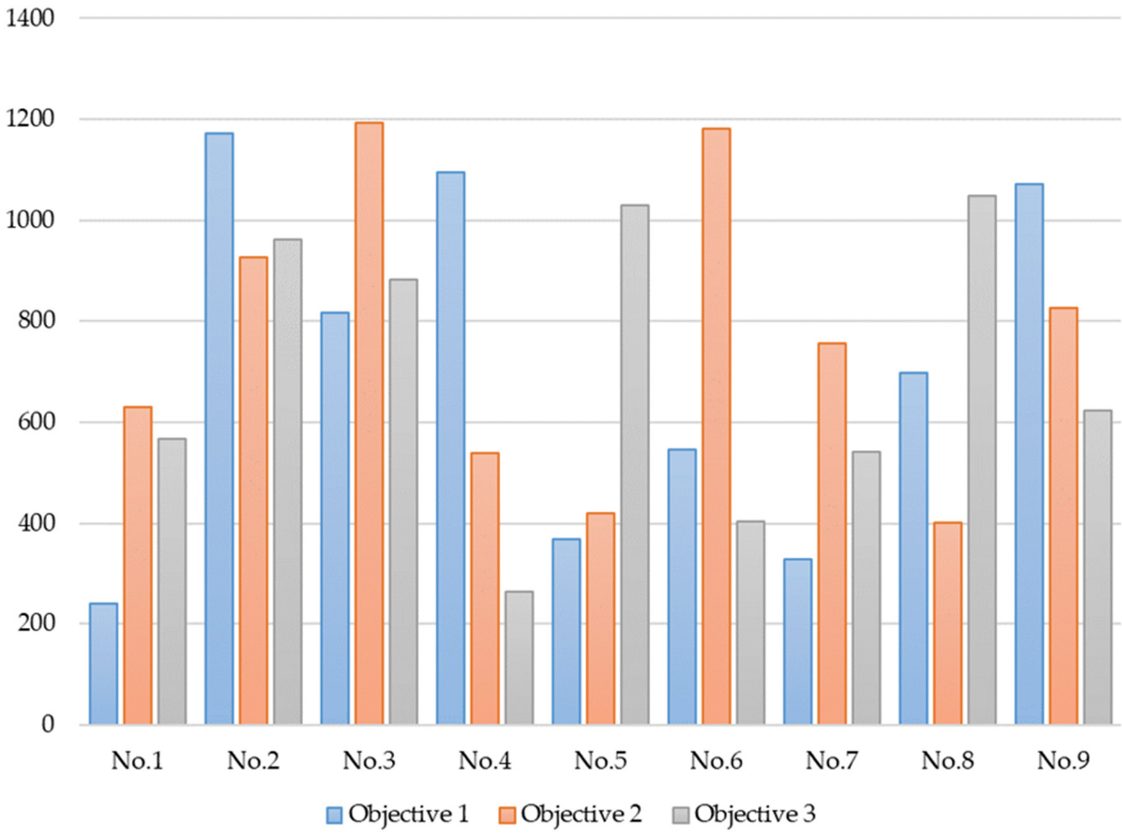

Figure 9. GA objective function value.

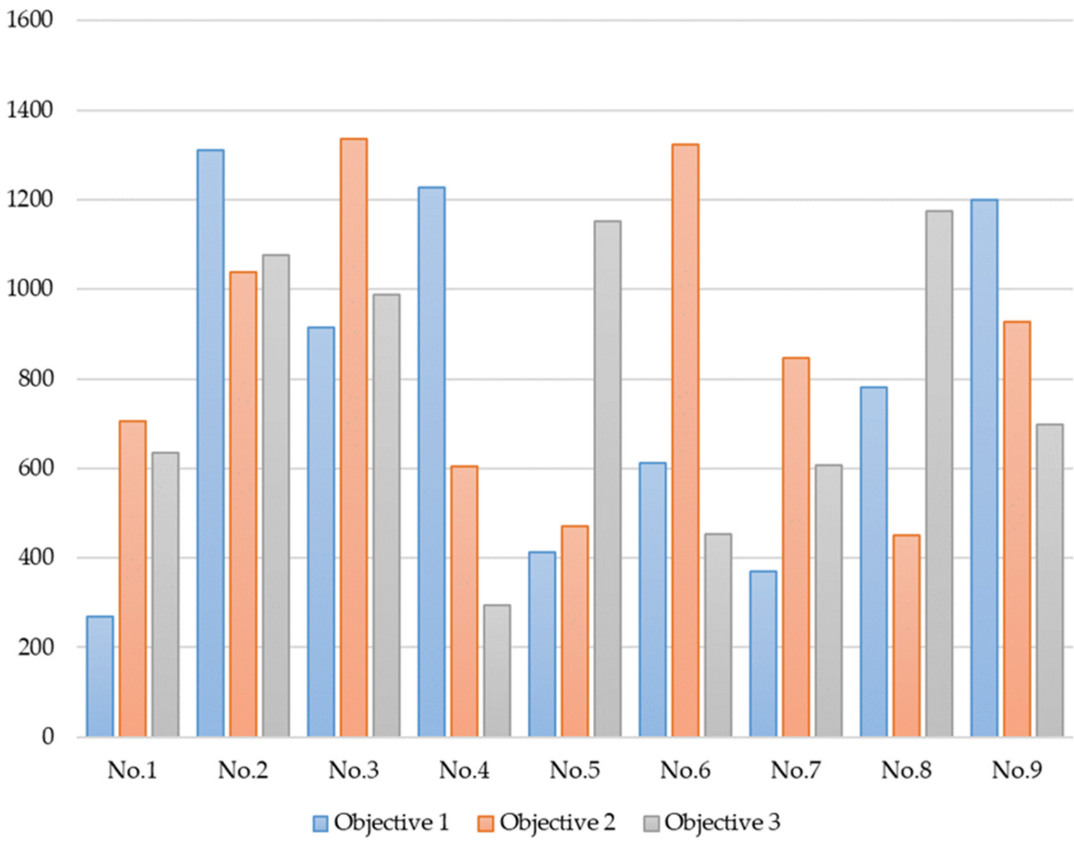

Figure 10. PSO objective function value.

This approach integrates forecasting and optimization to provide a practical framework that employs the concept of health equity to prioritize patients with the assistance of medical specialists based on their health conditions and need for blood transfusions in their treatments. Equations (16)–(20) incorporate prioritization into the model. Furthermore, the cross-matching technique has been employed in the model (Equations (21)–(23)) to mitigate blood shortages in alignment with the model's substitution and resiliency capacities.

The results of the model, coupled with the absence of a comprehensive model in the literature applicable to a transfusion organization like IBTO, indicate that the proposed model can be effectively implemented in real-world scenarios. It is capable of performing adequately to address the daily requirements of the BSC and assist decision-makers in managing blood inventory and distribution optimally under various real-life conditions.

5. Conclusions

This work presents a practical approach to blood supply chains and other perishable supply chains with similar characteristics. The model's effectiveness depends on its capacity to incorporate any factors that may affect the blood supply network. Implementing this methodology will boost the sustainability of the blood supply chain and promote the principle of health equity. Health equality principles will assist the healthcare systems in prioritizing medical care services for those in greater need. In recent years, the COVID-19 pandemic compelled healthcare systems worldwide to prioritize patients; this study applies the same principle to establish a sustainable approach capable of addressing any situations.

In contrast to prior studies, the integration of forecasting with optimization, as demonstrated in the research, renders the model realistic and enables blood transfusion experts and decision-makers to depend on it as a trustworthy tool. Whether from local centers or supporting entities, the model will efficiently manage blood demand and supply, addressing both surpluses and shortages.

The model's numerous variables render it NP-hard, preventing exact approaches from producing findings in a timely manner. Therefore, we employ a genetic algorithm to solve the model. The comparative analysis demonstrated the better efficacy of GA compared to PSO in tackling multiple objectives across various circumstances. The model offers a strong decision-support framework; however, it is essential to solve its limits and include more dynamic and scalable components for wider use.

The results indicate that utilizing weights to convert a multi-objective problem into a singular one allows for the effective management of each objective function based on unique conditions. This feature enables the system to self-adjust for optimal performance in any setting, whether normal or crisis, unlike previous models.

The current study encountered certain limitations and suggests potential avenues for future research. The integration of dynamic transportation routing represents a significant avenue for future research development. This study excludes staff scheduling and resource planning because it focuses on blood distribution. The integration of blood donor and staff scheduling, along with resource planning, would enhance the model presented in this research document. Furthermore, this study assumes that all BCS will be operational under all conditions; incorporating and simulating the impact of the unavailability of specific infrastructure due to interruptions would be a suitable approach. Figure 4 illustrates that most IBTO centers in Tehran are located in the eastern and southern regions of the district. This offers a beneficial opportunity to create new centers in the western region and assess the impact of these centers on the proposed model. This paper advises researchers to use advanced methodologies and machine learning techniques for the forecasting stage. Researchers may improve solving time through the application of hybrid methodologies or by utilizing solvers available in Python libraries. To improve the model's adaptability to real-life scenarios, researchers may integrate concepts such as the Internet of Things (IoT) as outlined in Industry 5.0. Researchers can enhance the realism of the model by utilizing interactive problem-solving methods and machine-learning techniques. The current model has no explicit implementation of environmental sustainability measures, which could align with the United Nations Sustainable Development Goals (SDGs). Future research may overcome these limitations by integrating real-time data, expanding the model to dynamic environments, and incorporating environmental and social factors into the optimization process.

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